

1-8 Figure P1-8 shows a plot of the net positive charge flowing in a wire versus time. Sketch the corresponding current during the same period of time.

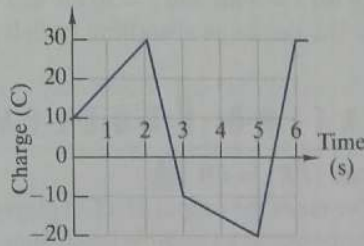


FIGURE P1-8

1-9 The net negative charge flowing through a device varies as $q(t) = 3t^2$ C. Find the current through the device at $t = 0$ s, $t = 0.5$ s, and $t = 1$ s.

1-10 A cell-phone charger outputs 9.6 V and is protected by a 50 mA fuse. A 1.5 W cell phone is connected to it to be charged. Will the fuse blow?

1-11 For $0 \leq t \leq 5$ s, the current through a device is $i(t) = 4t$ A. For $5 < t \leq 10$ s, the current is $i(t) = 40 - 4t$ A, and $i(t) = 0$ A for $t > 10$ s. Sketch $i(t)$ versus time and find the total charge flowing through the device between $t = 0$ s and $t = 10$ s.

1-12 The charge flowing through a device is $q(t) = 1 - e^{-1000t}$ μ C. How long will it take the current to reach 200 μ A?

1-13 The 12-V automobile battery in Figure P1-13 has an output capacity of 100 ampere-hours (Ah) when connected to a head lamp that absorbs 200 watts of power. The car engine is not running and therefore not charging the battery. Assume the battery voltage remains constant.

- Find the current supplied by the battery and determine how long can the battery power the headlight.
- A 100 W device is connected through the utility port. How long can the battery power both the headlight and the device?

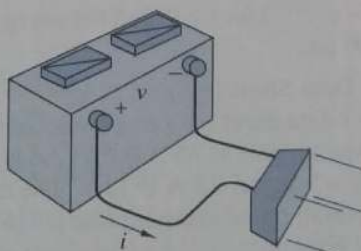


FIGURE P1-13

1-14 An incandescent lamp absorbs 100 W when connected to a 120-V source. A energy-efficient compact fluorescent lamp (CFL) producing the same amount of light absorbs 16 W when connected to the same source.

How much cheaper is it to operate the CFL versus the incandescent bulb over 1,000 hours when electricity costs 7.8 cents/kWh?

1-15 The current through a device is zero for $t < 0$ and is $i(t) = 3e^{-2t}$ A for $t \geq 0$. Find the charge $q(t)$ flowing through the device for $t \geq 0$.

1-16 A string of holiday lights is protected by a 5-A fuse and has 25 bulbs, each of which is rated at 7 W. How many strings can be connected end-to-end across a 120 V circuit without blowing a fuse?

1-17 When illuminated the i - v relationship for a photocell is $i = e^{-v} - 10$ A. For $v = -2, 2$ and 3 V find the device power and state whether it is absorbing or delivering power.

1-18 A new 6 V Alkaline lantern battery delivers 237.5 kJ of energy during its lifetime. How long will the battery last in an application that draws 15 mA continuously. Assume the battery voltage is constant.

1-19 The maximum power a device can dissipate is 0.25 W. Determine the maximum current allowed by the device power rating when the voltage is 9 V.

1-20 Traffic lights are being converted from incandescent bulbs to LED arrays to save operating and maintenance costs. Typically each incandescent light uses three 100-W bulbs, one for each color R, Y, G. A competing LED array consists of 61 LEDs with each LED requiring 9 V and drawing 20 mA of current. There are three arrays per light - R, Y, G. A small city has 1560 traffic signals. Since one light is always on 24/7, how much can a city save in one year if the city buys their electricity at 7.2¢ per kWh?

1-21 Two electrical devices are connected as shown in Figure P1-21. Using the reference marks shown in the figure, find the power transferred and state whether the power is transferred from A to B or B to A when

- $v = +11$ V and $i = -1.1$ A
- $v = +80$ V and $i = +20$ mA
- $v = -120$ V and $i = -12$ mA
- $v = -1.5$ V and $i = -600$ μ A

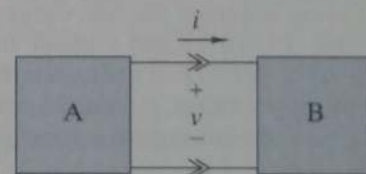


FIGURE P1-21

1-22 Figure P1-22 shows an electric circuit with a voltage and a current variable assigned to each of the six devices. The device voltages and currents are observed to be

	v (V)	i (A)		v (V)	i (A)
Device 1	15	-1	Device 4	-10	-1
Device 2	5	1	Device 5	20	-3
Device 3	10	2	Device 6	20	2

PROBLEMS

OBJECTIVE 2-1 ELEMENT CONSTRAINTS (SECT. 2-1)

Given a two-terminal element with one or more electrical variables specified, use the element i - v constraint to find the magnitude and direction of the unknown variables.

See Examples 2-1, 2-2, 2-3, and 2-4 and Exercises 2-1, 2-2, and 2-3.

2-1 The current through a 56-k Ω resistor is 2.2 mA. Find the voltage across the resistor.

2-2 The voltage across a particular resistor is 6.23 V and the current is 2.75 mA. What is the actual resistance of the resistor? Using the inside back cover, what is the likely standard value of the resistor?

2-3 A 100-k Ω resistor dissipates 100 mW. Find the current through the resistor.

2-4 The conductance of a particular resistor is 0.5 mS. Find the current through the resistor when connected across a 5-V source.

2-5 In Figure P2-5 the resistor dissipates 25 mW. Find R_x .

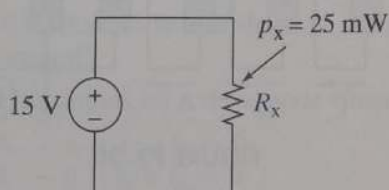


FIGURE P2-5

2-6 In Figure P2-6 find R_x and the power delivered to the resistor.

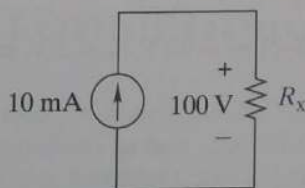


FIGURE P2-6

2-7 A resistor found in the lab has three orange stripes followed by a gold stripe. An ohmmeter measures its resistance as 34.9 k Ω . Is the resistor properly color coded? (See inside back cover for color code.)

2-8 The i - v characteristic of a nonlinear resistor is $v = 82i + 0.18i^3$.

(a) Calculate v and p for $i = \pm 0.5, \pm 1, \pm 2, \pm 5$, and ± 10 A.

(b) Find the maximum error in v when the device is treated as an 82- Ω linear resistance on the range $|i| < 0.5$ A.

2-9 A 100-k Ω resistor has a power rating of 0.125 W. Find the maximum voltage that can be applied to the resistor.

2-10 A certain type of film resistor is available with resistance values between 10 Ω and 100 M Ω . The maximum ratings for all resistors of this type are 500 V and 0.25 W. Show that the voltage rating is the controlling limit for $R > 1$ M Ω , and that the power rating is the controlling limit when $R < 1$ M Ω .

2-11 Figure P2-11 shows the circuit symbol for a class of two-terminal devices called diodes. The i - v relationship for a specific pn junction diode is $i = 2 \times 10^{-16} (e^{40v} - 1)$ A.

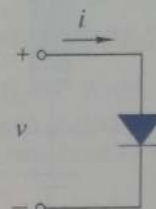


FIGURE P2-11

(a) Use this equation to find i and p for $v = 0, \pm 0.1, \pm 0.2, \pm 0.4$, and ± 0.8 V. Use these data to plot the i - v characteristic of the element.

(b) Is the diode linear or nonlinear, bilateral or nonbilateral, and active or passive?

(c) Use the diode model to predict i and p for $v = 5$ V. Do you think the model applies to voltages in this range? Explain.

(d) Repeat (c) for $v = -5$ V.

2-12 A thermistor is a temperature-sensing element composed of a semiconductor material that exhibits a large change in resistance proportional to a small change in temperature. A particular thermistor has a resistance of 5 k Ω at 25 $^{\circ}$ C. Its resistance is 340 Ω at 100 $^{\circ}$ C. Assuming a straight-line relationship between these two values, at what temperature will the thermistor's resistance equal 1 k Ω ?

OBJECTIVE 2-2 CONNECTION CONSTRAINTS (SECT. 2-2)

Given a circuit composed of two-terminal elements:

(a) Identify nodes and loops in the circuit

(b) Identify elements connected in series and in parallel.

(c) Use Kirchhoff's laws (KCL and KVL) to find selected signal variables.

See Examples 2-5, 2-6, and 2-7 and Exercises 2-4, 2-5, 2-6, 2-7, and 2-8

2-13 In Figure P2-13 $i_2 = -5$ A and $i_3 = 2$ A. Find i_1 and i_4 .

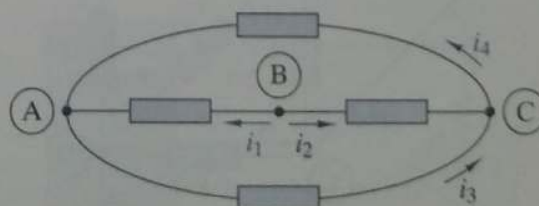


FIGURE P2-13

2-14 In Figure P2-14 $v_1 = 3$ V and $v_3 = 5$ V. Find v_2 , v_4 and v_5 .

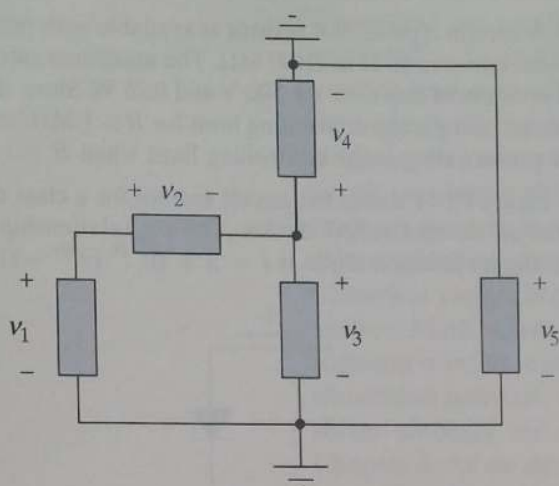


FIGURE P2-14

2-15 For the circuit in Figure P2-15:

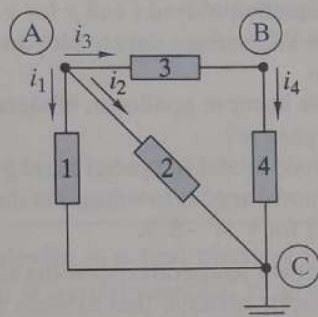


FIGURE P2-15

- Identify the nodes and at least two loops.
- Identify any elements connected in series or in parallel.
- Write KCL and KVL connection equations for the circuit.

2-16 In Figure P2-15 $i_2 = -20$ mA and $i_4 = 10$ mA. Find i_1 and i_3 .

2-17 For the circuit in Figure P2-17:

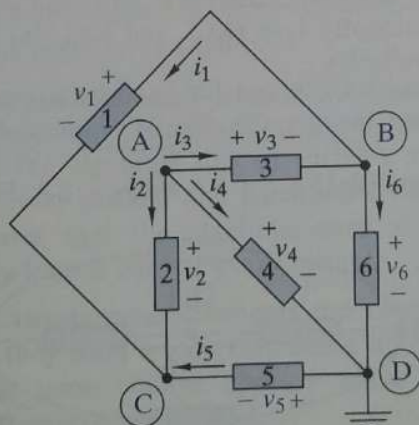


FIGURE P2-17

- Identify the nodes and at least three loops in the circuit.
- Identify any elements connected in series or in parallel.
- Write KCL and KVL connection equations for the circuit.

2-18 In Figure P2-17 $v_2 = 10$ V, $v_3 = -10$ V, and $v_4 = 3$ V. Find v_1 , v_5 , and v_6 .

2-19 In many circuits the ground is often the metal case that houses the circuit. Occasionally a failure occurs whereby a wire connected to a particular node touches the case causing that node to become connected to ground. Suppose that in Figure P2-17 Node B accidentally touches ground. How would that affect the voltages found in problem 2-18?

2-20 The circuit in Figure P2-20 is organized around the three signal lines A, B, and C.

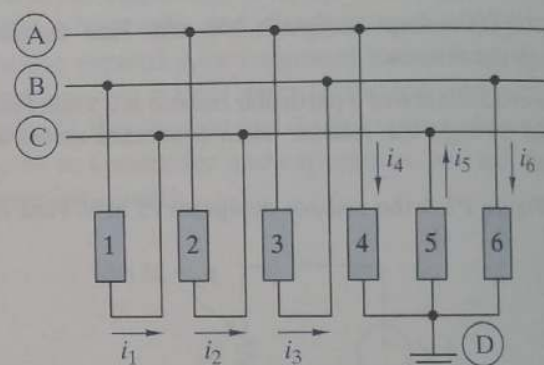


FIGURE P2-20

- Identify the nodes and at least three loops in the circuit.
- Write KCL connection equations for the circuit.
- If $i_1 = -20$ mA, $i_2 = -12$ mA, and $i_3 = 50$ mA, find i_4 , i_5 , and i_6 .
- Show that the circuit in Figure P2-20 is identical to that in Figure P2-17.

2-21 In Figure P2-21 $v_2 = 10$ V, $v_4 = 5$ V, and $v_5 = 15$ V. Find v_1 , v_3 , and v_6 .

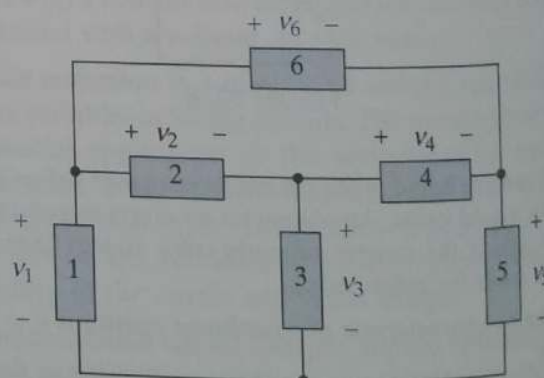


FIGURE P2-21

2-22 In Figure P2-22 $i_1 = 25$ mA, $i_2 = 10$ mA, and $i_3 = -15$ mA. Find i_4 and i_5 .

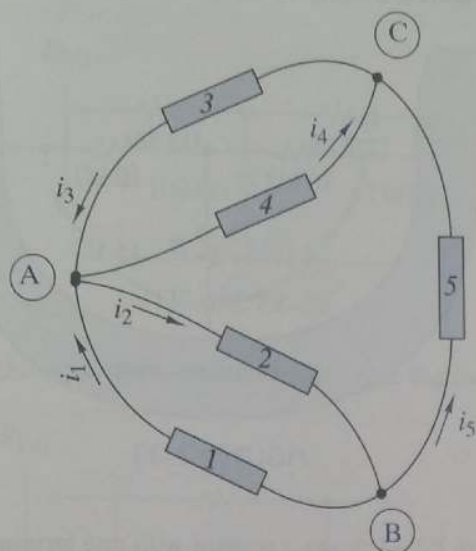


FIGURE P2-22

2-23 (a) Use the passive sign convention to assign voltage variables consistent with the currents in Figure P2-22. Write three KVL connection equations using these voltage variables.

(b) If $v_4 = 0$ V, what can be said about the voltages across all the other elements?

2-24 The KCL equations for a three-node circuit are:

$$\text{Node A} \quad -i_1 + i_2 - i_4 = 0$$

$$\text{Node B} \quad -i_1 - i_3 + i_5 = 0$$

$$\text{Node C} \quad i_1 + i_3 + i_4 - i_5 = 0$$

Draw the circuit diagram and indicate the reference directions for the element currents.

OBJECTIVE 2-3 COMBINED CONSTRAINTS (SECT. 2-3)

Given a linear resistance circuit, use the element constraints and connection constraints to find selected signal variables. See Examples 2-8, 2-9, 2-10, 2-11, and 2-12 and Exercise 2-9, 2-10, 2-11, 2-12, and 2-13.

2-25 Find v_x and i_x in Figure P2-25.

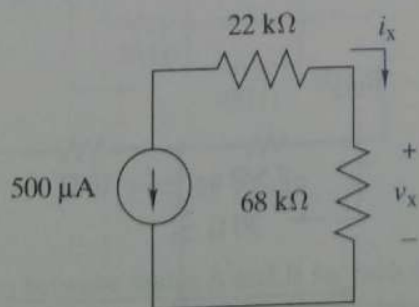


FIGURE P2-25

2-26 Find v_x and i_x in Figure P2-26.

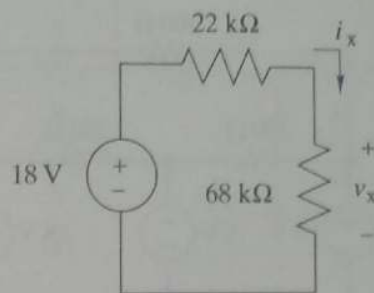


FIGURE P2-26

2-27 Find v_x and i_x in Figure P2-27. Compare the results of your answers with those in problem 2-26. What effect did adding the 33-kΩ resistor have on the overall circuit? Why isn't i_y zero?

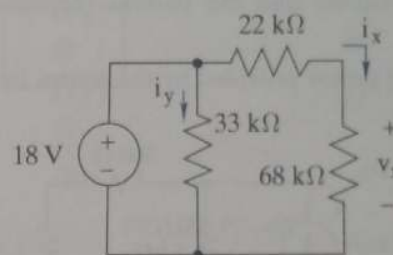


FIGURE P2-27

2-28 A modeler wants to light his model building using miniature grain-of-wheat light bulbs connected in parallel as shown in Figure P2-28. He uses two 1.5-V "C-cells" to power his lights. He wants to use as many lights as possible, but wants to limit his current drain to 500 μA to preserve the batteries. If each light has a resistance of 36 kΩ, how many lights can he install and still be under his current limit?

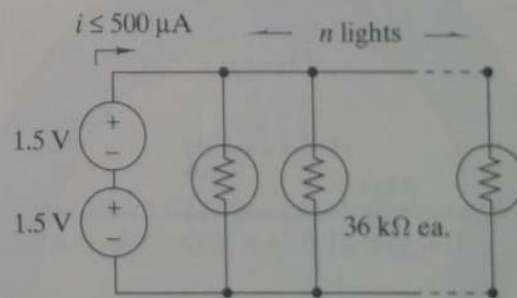


FIGURE P2-28

2-29 Find v_x and i_x in Figure P2-29.

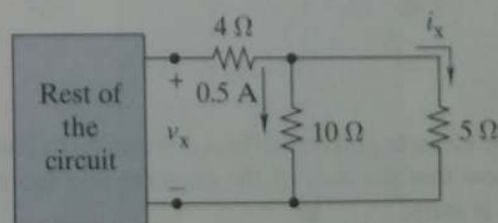


FIGURE P2-29